Chao Zhou (周超)

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Citizenship: Chinese

Research interests

(efficient) machine learning, efficient large model finetuning, neural network architecture design, generative model, image and signal processing

Education

University College London (UCL) London, UK
PhD in electronic and electrical engineering Sep.2018 – Sep.2023
Supervisor: Professor Miguel Rodrigues

University of Electronic S&T of China (UESTC) Chengdu, China
Master in Communication and Information System Sep 2015 – Jun 2018
Mentor: Professor Xuesong Jonathan Tan

Wuhan University of Technology (WHUT) Wuhan, China
Bachelor in Communication Engineering Sep 2011 – Jun 2015
Mentor: Professor Kewen Liu

Industry experience

Huawei Technologies Co., Ltd. Shenzhen, China
Senior R&D Engineer Jan 2024-Apr 2024

Huawei Technologies Research & Development (UK) Ltd. London, UK
Research Intern Jun 2022-Jan 2023

Research experience

Relational Machine Learning Group, CISPA Helmholtz Center for Information Security, Saarbrücken, Germany **Postdoc**
Supervisor: Dr. Rebekka Burkholz May.2024 – present

Project: SPARE-ML

Description: This project is funded by the European Research Council (ERC) and aims to combine ideas from statistical physics and machine learning to explore the potential of deep learning models at a larger scale. In large-model post-training, especially supervised fine-tuning (SFT), we observed two phenomena: first, parameters that receive larger gradient updates tend to have update magnitudes inversely proportional to their absolute values, and this inverse relationship is more pronounced during fine-tuning than during pretraining; second, full-parameter fine-tuning can easily trigger catastrophic forgetting, causing the model to rapidly lose knowledge acquired during pretraining and thereby limiting its ability in continual learning scenarios. Based on these observations, we propose a new efficient fine-tuning method, NanoAdam. When fine-tuning large models, this method updates only a subset of parameters with small weight magnitudes while freezing those with large weights. To preserve the model's generalization ability and avoid over-reliance on any fixed parameter subset, we reselect the "small-weight parameter subset" every training steps. Experimental results show that, compared with full-parameter fine-tuning and LoRA, our method achieves better generalization on downstream tasks. Theoretically, its memory footprint is only about 1.5% of traditional Adam fine-tuning. In continual learning tasks, full-parameter fine-tuning with Adam can reduce performance on the previous task by up to 83%, whereas our method reduces it by only 12%.

Project Achievements: The project has produced 5 papers in top AI conferences, including ICML, NeurIPS, ICLR, and CVPR.

Information, Inference and Machine Learning (I2ML) lab, UCL, London, UK, **PhD**

Mentor: Professor Miguel Rodrigues Sep.2018 – Sep.2023

Project: ARTICT

Description: Brings together researchers in Information and Communication Technology (ICT) and Heritage Science to foster the exchange of ideas and expertise—co-creating novel automated signal analysis and processing tools that integrate MA-XRF (macro X-ray fluorescence) and HSI (hyperspectral imaging) data to support technical research, conservation, and preservation of artworks. With the advancement of machine learning, hyperspectral image (HSI) unmixing has increasingly adopted learning-based approaches. However, without proper guidance, these methods cannot guarantee physically meaningful unmixing results. This paper proposes an unsupervised framework inspired by Deep Image Prior (DIP) that works simultaneously for linear and nonlinear blind unmixing models. The framework consists of three modules: 1. EDIP (Endmember DIP) - endmember estimation module 2. ADIP (Abundance DIP) - abundance estimation module 3. MM (Mixing Module) - reconstructs HSI observations based on the assumed unmixing model EDIP and ADIP generate endmembers and abundances respectively, while MM reconstructs observations. We designed a composite loss function applicable to both linear and nonlinear unmixing models to produce physically meaningful results. Additionally, for nonlinear mixing scenarios, we propose an adaptive loss weighting strategy for better unmixing performance. Extensive experiments on synthetic and real datasets demonstrate that our method outperforms state-of-the-art unmixing algorithms.

Project Responsibilities: Participate as a PhD student in the research project; Research HSI imaging analysis techniques based on machine learning.

Project Achievements: 3 papers in top SCI journals (IEEE TGRS, TNNLS, TIP), 4 EI conference papers (IEEE ICASSP, ICIP)

Publications

1. MA Lodhi, **C. Zhou**, R Burkholz, ”RoSA: Rotational Sparse Adaptation for Memory-Efficient Fine-Tuning”, in NeurIPS 2026.

2. W. Pu, Y. Song, **C. Zhou** and et al., ”DIAI-net: An Integrated Network for Deep SAR Imaging and Autofocusing,” in IEEE Transactions on Aerospace and Electronic Systems, doi: 10.1109/TAES.2026.3701857.

3. L Chaves, **C. Zhou**, R Burkholz, E Valle, S Avila. ”Bridging Domains through Subspace-Aware Model Merging”, accepted by CVPR 2026.

4. T Jacobs, **C. Zhou**, R Burkholz. ”Never Saddle: Reparameterized Steepest Descent as Mirror Flow.”, accepted by ICLR 2026.

5. **C. Zhou**, T Jacobs, A Gadhikar, R Burkholz. ”Pay Attention to Small Weights.” accepted by NeurIPS 2025.

6. A Gadhikar, T Jacobs, **C. Zhou**, R Burkholz. (2025). Sign-In to the Lottery: Reparameterizing Sparse Training From Scratch. accepted by NeurIPS 2025.

7. T Jacobs, **C. Zhou**, R Burkholz. ”Mirror, Mirror of the Flow: How Does Regularization Shape Implicit Bias?.” arXiv preprint arXiv:2504.12883, accepted by ICML 2025.

8. **C. Zhou**, Pu Wei, and M. R. D. Rodrigues. ”Neural Network for Blind Unmixing: a novel MatrixConv Unmixing (MCU) Approach.” arXiv preprint arXiv:2503.08745 (2025).

9. **C. Zhou** and M. R. D. Rodrigues, ”Hyperspectral Blind Unmixing Using a Double Deep Image Prior,” in IEEE Transactions on Neural Networks and Learning Systems, doi: 10.1109/TNNLS.2023.3294714.

10. **C. Zhou**, Z. Lyu and M. R. D. Rodrigues, ”BITS-Net: Blind Image Transparency Separation Network,” 2023 IEEE International Conference on Image Processing (ICIP), Kuala Lumpur, Malaysia, 2023, pp. 375-379, doi: 10.1109/ICIP49359.2023.10222918.

11. W. Pu, B Sober, N. Daly, **C. Zhou**, Z. Sabetsarvestani, C. Higgitt, I. Daubechies, and M. R. D. Rodrigues, ‘‘Image Separation with Side Information: A Connected Auto-Encoders Based Approach,” in IEEE Transactions on Image Processing, vol. 32, pp. 2931-2946, 2023, doi: 10.1109/TIP.2023.3275872.

12. **C. Zhou** and M. R. D. Rodrigues, ”Blind Unmixing Using A Double Deep Image Prior,” ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Singapore, Singapore, 2022, pp. 1665- 1669, doi: 10.1109/ICASSP43922.2022.9747545.

13. W. Pu, **C. Zhou**, Y. C. Eldar, and M. R. D. Rodrigues. ”Robust lEarned Shrinkage-Thresholding (REST): Robust unrolling for sparse recover.” arXiv preprint arXiv:2110.10391 (2021).

14. W. Pu, **C. Zhou**, Y. C. Eldar and M. R. D. Rodrigues, "REST: Robust lEarned Shrinkage-Thresholding Network Taming Inverse Problems with Model Mismatch," ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, ON, Canada, 2021, pp. 2885-2889, doi: 10.1109/ICASSP39728.2021.9414141.

15. **C. Zhou** and M. R. D. Rodrigues, "An ADMM Based Network for Hyperspectral Unmixing Tasks," ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, ON, Canada, 2021, pp. 1870-1874, doi: 10.1109/ICASSP39728.2021.9414555.

16. X. J. Tan, **C. Zhou** and J. Chen, "Symmetric Channel Hopping for Blind Rendezvous in Cognitive Radio Networks Based on Union of Disjoint Difference Sets," in IEEE Transactions on Vehicular Technology, vol. 66, no. 11, pp. 10233-10248, Nov. 2017, doi: 10.1109/TVT.2017.2726352.

Services

Conference Reviewer: EUSIPCO, ICLR, ICML, CVPR, ECCV.
Journal Reviewer: IEEE TGRS, IEEE JSTSP, IEEE TIT, IEEE Access.
Area Chairs: CPAL.

Teaching experience

Teaching assistant, UCL Consultants Limited and Turing Innovations Ltd 2020-2023

Machine Learning Masterclass for Defence Science and Technology Laboratory (DSTL), UK

Duties: Mini-project; Answer questions re. mini-project; Weekly programming Assignment/Tasks; Answer questions re. weekly programming assignment/tasks; Answer moodle questions re above; Quizzes; Reading material; Maintain/Fill-in moodle pages.

Post Graduate Teaching Assistant, Dept. of Electronic and Electrical Engineering at UCL Fall 2020 & Fall 2021

ELEC0134: Applied Machine Learning Systems

Duties: Mini-project; Answer questions re. mini-project; Weekly programming Assignment/Tasks; Answer questions re. weekly programming assignment/tasks; Answer moodle questions re above; Quizzes; Reading material; Maintain/Fill-in moodle pages.

Post Graduate Teaching Assistant, Dept. of Electronic and Electrical Engineering at UCL Fall 2019

ELEC0136: Data Acquisition and Processing Systems

The module will cover aspects of data acquisition, data visualization, data cleaning, data analysis and processing.

Honors and scholarships

The First-class National Scholarship of Graduate Students (UESTC) 2017
The Seventh Place of College Fitness of Mianyang Fitness Bodybuilding Championships 2016
The Outstanding Graduate Student (UESTC) 2016
The First-class National Scholarship of Graduate Students (UESTC) 2016
The First-class National Scholarship of Graduate Students (UESTC) 2015
Advanced Individuals of Innovation and Entrepreneurship for work of Communist Youth League (WHUT) 2014
The National Endeavor Fellowship of Undergraduate Students, (WHUT) 2013
The First-Class Prize of National Undergraduate Electronics Design Contest (WHUT) 2013
The University Outstanding Student Leaders (WHUT) 2013
The Second-Class Prize of the Sixth Electric and Electronic Innovation Design Contest (WHUT) 2013
The School Outstanding Student (WHUT) 2012

Skills

Programming
Proficient in: python, pytorch, matlab.
Familiar with: C, Java.

Languages
Chinese, English

Other interests

Bodybuilding.